

COMPREHENSIVE CALCULUS: DIFFERENTIATION & INTEGRATION

1. Foundations: Limits & Continuity (Recap)

Calculus rests upon the behavior of functions near a point. A function is continuous if the limit as x approaches c equals $f(c)$.

2. Differential Calculus

The Derivative

The derivative $f'(x)$ measures the instantaneous rate of change of y with respect to x , geometrically defined as the slope of the tangent line.

$$f'(x) = \lim_{h \rightarrow 0} [f(x+h) - f(x)] / h$$

Key Differentiation Rules:

- Power Rule: $d/dx(x^n) = nx^{(n-1)}$

- Product Rule: $(uv)' = u'v + uv'$
 - Quotient Rule: $(u/v)' = (u'v - uv') / v^2$
 - Chain Rule: $d/dx(f(g(x))) = f'(g(x)) * g'(x)$
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3. Long Answer: Differentiation Problems

Problem: Differentiate $f(x) = \sin(x^2) * e^{(3x)}$.

Solution:

Using the Product Rule: $u = \sin(x^2)$, $v = e^{(3x)}$. $u' = \cos(x^2) * 2x$ (by Chain Rule). $v' = e^{(3x)} * 3$. $f'(x) = [2x * \cos(x^2)] * e^{(3x)} + \sin(x^2) * [3 * e^{(3x)}]$.
 $f'(x) = e^{(3x)} * [2x * \cos(x^2) + 3 * \sin(x^2)]$.

4. Integral Calculus

Fundamental Theorem of Calculus

Integration is the reversal of differentiation and represents the signed area under a curve.

$$\int_a^b f(x) \, dx = F(b) - F(a)$$

Integration Techniques:

- **Substitution:** Transforming $\int f(g(x))g'(x)dx$ into $\int f(u)du$ where $u = g(x)$.
- **Integration by Parts:** $\int u \, dv = uv - \int v \, du$.
- **Partial Fractions:** Decomposing rational functions into simpler fractions.

5. Long Answer: Integration Problems

Problem: Evaluate $\int x * \ln(x) \, dx$.

Solution (By Parts):

Let $u = \ln(x)$ and $dv = x \, dx$.

Then $du = (1/x) \, dx$ and $v = x^2 / 2$.

$\int u \, dv = (\ln(x) * x^2 / 2) - \int (x^2 / 2) * (1/x) \, dx$.

$= (x^2 / 2) * \ln(x) - (1/2) * \int x \, dx$.

$= (x^2 / 2) * \ln(x) - (1/2) * (x^2 / 2) + C$.

$= (x^2 / 2) * \ln(x) - x^2 / 4 + C$.

6. Advanced Applications

Differential equations, optimization (finding maxima and minima), and volume of solids of revolution are the primary applications of calculus in physical sciences.

7. Summary Table of Derivatives & Integrals

|(Standard integrals and derivatives of trig, exp, and log functions...)